

Ching-Han Cheng

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OBJECTIVE

Seeking an R&D Substitute Service position in Linux kernel or system software, building on hands-on projects in OS internals, kernel modules, and low-level debugging tools.

EDUCATION

- **National Yang Ming Chiao Tung University (NYCU)** Feb. 2024 - Jul. 2026
Institute of Network Engineering
◦ Advisor: [Li-Hsing Yen](#)
◦ Thesis Title: Decentralized Workload Offloading for Carbon Reduction and Queue Stability in a Hybrid Cloud-Edge Environment [🔗]
◦ GPA: 4.15/4.30
- **National Sun Yat-Sen University (NSYSU)** Sep. 2020 - Jan. 2024
Department of Computer Science and Engineering
◦ Ranking: 5/66
◦ GPA: 4.08/4.30

PROJECTS

- **xv6-riscv Kernel: [MIT 6.1810 Operating Systems Labs] (Self-Study, In Progress)** Jul. 2026 - Present
Tools: [C, RISC-V, GDB, QEMU] [🔗]
◦ Independently completed MIT 6.1810's first 5 kernel labs, then resolved cross-subsystem merge conflicts (page tables, copy-on-write) to unify them into one regression-free kernel
◦ Designed a two-tier (4KB page / 2MB superpage) buddy-style physical memory allocator with automatic split/coalesce, and extended the *brk* syscall with eager/lazy and normal/super allocation flags
- **Single Instruction Debugger: [NYCU Advanced Programming in the UNIX Environment]** Spring 2025
Tools: [C++, Capstone, ptrace] [🔗]
◦ Built a ptrace-based x86-64 instruction-level debugger (step, continue, breakpoints, memory patching, syscall tracing, Capstone disassembly) for static, dynamic and PIE binaries
◦ Redesigned breakpoint handling to defer restoring the original byte until resume (restore, single-step, re-insert), fixing breakpoints on jump instructions
- **Remote Working Ground Server: [NYCU Network Programming]** Spring 2025
Tools: [C++] [🔗] Project 2, 3]
◦ Built a shared shell interpreter with numbered pipes and redirection, featuring built-in chat and firewall
◦ Project 2: single process multiplexing up to 40 clients with `select()`, per-user environments, user-to-user pipes
◦ Project 3: one process per client, sharing user state through POSIX shared memory, signals, and FIFO user pipes
- **AES Linux Kernel Module: [NYCU Advanced Programming in the UNIX Environment]** Spring 2025
Tools: [C, QEMU, Linux Kernel] [🔗]
◦ Implemented a character-device kernel module providing AES-ECB via `read/write/ioctl`, with per-fd state
◦ Traced cross-process data corruption to a double-free inherited from the sample code, using SLUB debug in QEMU

TEACHING ASSISTANT EXPERIENCE

- **Introduction to Network Programming** Fall 2025
NYCU
◦ Pilot-solved labs before release, then advised the instructor on unclear requirements and hints to add
◦ Refactored and customized shell-based automated testing scripts using *tmux* to accelerate student assignment testing and streamline grading workflows [🔗 Demo script: combined]
[🔗 Demo script: split]

HONORS AND AWARDS

- **Excellent Student Awards (× 2)** Fall 2020, Spring 2021
NSYSU
◦ Top 5% of the department
- **Excellent Performance in Professional Courses (× 3)** Fall 2021, Spring 2022, Fall 2022
NSYSU

CERTIFICATIONS

- TOEIC: 805 points Aug. 2025
- Collegiate Programming Examination (CPE): Top 0.2% (4th/1,814), solving 6 out of 7 problems May 2022